Print guide

DIY CNC epoxy-granite mill · October 2026

Every printed part for the moulds, the castings and the machine, cut to fit a Bambu A1 mini (180 mm cube; no piece is over 170 mm). The STL files are in `print/`, one folder per casting, with the list in `print/manifest.csv`. They are written by `model/build_cnc.py` on every run. How the moulds go together is in the mould guide.

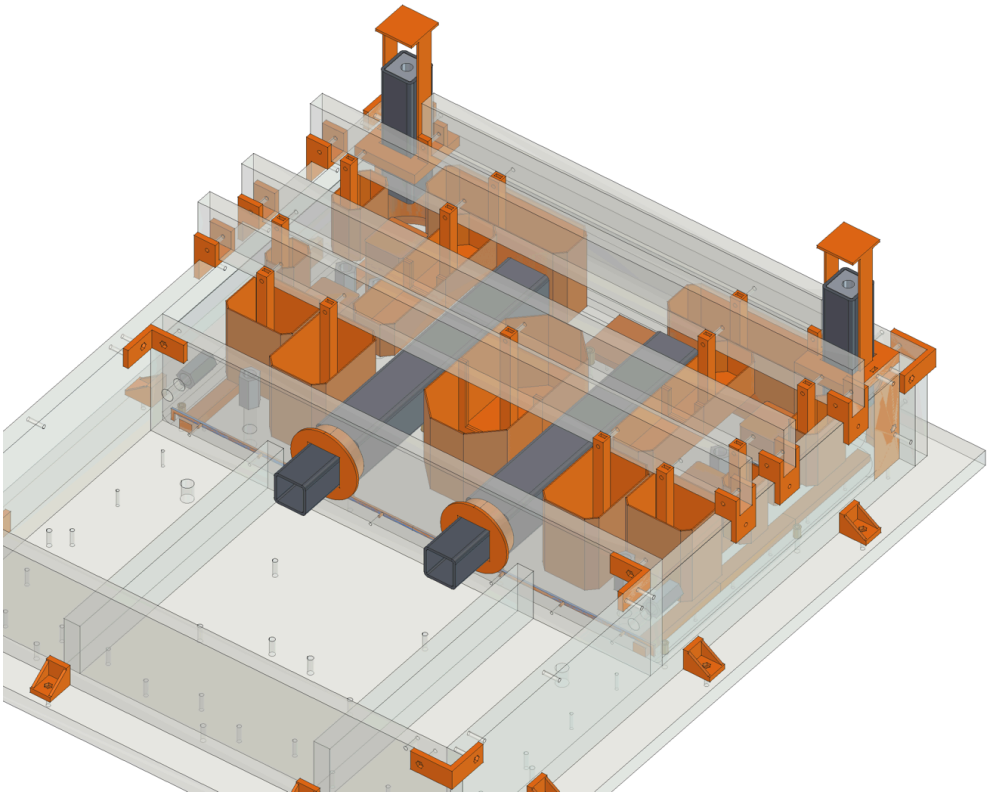
About 186 prints, roughly 3.7 kg of PETG. The filament figures are rough estimates.

Settings

- **PETG for everything.** Thick sections of curing epoxy get warm, and PLA softens at about 55 °C.
- 0.4 mm nozzle, 0.2 mm layers. Every file is already turned the way it prints best, without supports.
- **Cores** (`core-*`): thin shells, 1.2 mm walls (three lines), so they print solid; the posts get 2 walls and 10 % infill. The sand inside holds the walls while casting.
- **Fillers, formers, plugs** (`filler-*`, `former-*`, `plug-*`): 2 walls, 10 % infill. They only have to hold their shape against the mix; the nut pockets have their own walls.
- **Mould kit** (`kit-*`: brackets, saddles, holders, bushings): 3 walls, 15 % infill. Print one, check it over a 19 mm offcut of your board, then the rest.
- **Liners and cast-in parts** (`liner-*`, `cast-in-*`): 3 walls, 40 % infill.
- **Fits.** Slip fits are 0.15 mm a side, nut pockets 0.15 mm over the nut, pins 3.0 mm into a 1/8 in hole. If your printer runs tight, print a bracket and a former first and adjust the horizontal expansion (XY compensation) rather than the files.

Joining the pieces

- **Nuts.** Every piece that bolts to a board has its nut built in. On cast faces the nut drops into a pocket in the face against the board and slides sideways under the bolt hole; on brackets it presses into a hex pocket. Fit the nuts before the piece goes on the board.
- **Parts cut to fit the bed** (`-partNofM`) have a peg on one cut face and a socket on the other: push them together, no glue. Each part has its own bolts or pins to the board.
- **Cores** cut in two have a bulkhead on each side of the cut with two M3 x 10 bolts through them.
- **Pins.** Thin formers (gasket-groove beads, the shallow end of a gutter) have 3 mm pins that push into 1/8 in holes. Nothing else holds them; the casting presses them onto the board.
- **Removable pieces** (formers, fillers, plugs, kit): 3-5 coats of paste wax and PVA like the boards. **Pieces that stay in** (cores, liners, cast-in parts): no wax; scuff and degrease them so the epoxy grips. Fill the cores with dry sand before the bridge goes on.



Foam cores instead of printed ones

The pocket cores are about 1.6 kg of the PETG. The rectangular ones are quick to cut from XPS foam board on a table saw or with a hot wire; wrap them in packing tape. The corners are chamfered 8-10 mm (30 mm on the rear centre core). Set `foam_cores=True` in `model/build_cnc.py` and re-run it: the cores drop out of `print/` and each gets just its printed post instead (about 8 g), which pushes 30 mm into a snug hole in the foam and bolts to the bridge like the printed core's. Sizes (outline x depth from the underside):

Casting	Core	Outline (mm)	Depth (mm)
base-front	centre	100 x 185	90
base-rear	centre	100 x 190	90
base-front	outboard-L	130 x 164	90
base-front	trough-band-L	101 x 55	40
base-front	draw-rod-L	37 x 255	40
base-front	foot-a-L	68 x 66	90
base-rear	foot-b-L	68 x 100	90
base-rear	foot-c-L	80 x 63	90
base-rear	knee-L	50 x 183	40
base-rear	outer-a-L	35 x 43	65
base-rear	outer-b-L	35 x 60	65
base-rear	rear-L	170 x 52	90
base-front	outboard-R	130 x 164	90

Casting	Core	Outline (mm)	Depth (mm)
base-front	trough-band-R	101 x 55	40
base-front	draw-rod-R	37 x 255	40
base-front	foot-a-R	68 x 66	90
base-rear	foot-b-R	68 x 100	90
base-rear	foot-c-R	80 x 63	90
base-rear	knee-R	50 x 183	40
base-rear	outer-a-R	35 x 43	65
base-rear	outer-b-R	35 x 60	65
base-rear	rear-R	170 x 52	90

Base mould kit (both halves)

Folder `print/base-box/`, about 368 g.

File	What	Qty	Size (mm)	Notes
`kit-corner-bracket.stl`	Mould kit (reused)	4	50 x 50 x 23	
`kit-floor-bracket.stl`	Mould kit (reused)	10	24 x 32 x 25	
`kit-partition-bracket.stl`	Mould kit (reused)	2	44 x 44 x 23	
`kit-saddle.stl`	Mould kit (reused)	2	53 x 31 x 25	The rear half needs six; four move over for the front half.
`kit-leg-holder.stl`	Mould kit (reused)	4	72 x 72 x 40	
`kit-leg-hanger.stl`	Mould kit (reused)	4	43 x 43 x 145	Drops over the leg end and rests on the holder: the leg hangs at its cast-in depth.

Base, rear half (cast first)

Folder `print/base-rear/`, about 1310 g.

File	What	Qty	Size (mm)	Notes
`former-gutter-L-part1of3.stl`	Former (removable)	1	15 x 124 x 12	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-L-part2of3.stl`	Former (removable)	1	15 x 124 x 10	
`former-gutter-L-part3of3.stl`	Former (removable)	1	15 x 120 x 9	

File	What	Qty	Size (mm)	Notes
`former-gutter-R-part1of3.stl`	Former (removable)	1	15 x 124 x 12	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-R-part2of3.stl`	Former (removable)	1	15 x 124 x 10	
`former-gutter-R-part3of3.stl`	Former (removable)	1	15 x 120 x 9	
`former-motor-relief.stl`	Former (removable)	1	72 x 60 x 14	Bolts to the floor from underneath (the base is cast top face down).
`filler-corner-L.stl`	Filler (removable)	1	40 x 40 x 110	Rounded corner; bolts to the floor and both walls.
`filler-corner-R.stl`	Filler (removable)	1	40 x 40 x 110	Rounded corner; bolts to the floor and both walls.
`former-split-gasket-groove-part1of4.stl`	Former (removable)	1	145 x 14 x 8	Pins into the partition.
`former-split-gasket-groove-part2of4.stl`	Former (removable)	1	145 x 4 x 8	
`former-split-gasket-groove-part3of4.stl`	Former (removable)	1	145 x 4 x 8	
`former-split-gasket-groove-part4of4.stl`	Former (removable)	1	145 x 14 x 8	
`core-centre-part1of2.stl`	Pocket core (shell, stays in)	1	100 x 86 x 90	Shell core, stays in. Bolts to the bridge by its post.
`core-centre-part2of2.stl`	Pocket core (shell, stays in)	1	100 x 104 x 141	
`core-foot-b-L.stl`	Pocket core (shell, stays in)	1	68 x 100 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-foot-c-L.stl`	Pocket core (shell, stays in)	1	80 x 63 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-knee-L-part1of2.stl`	Pocket core (shell, stays in)	1	50 x 92 x 40	Shell core, stays in. Bolts to the bridge by its post.
`core-knee-L-part2of2.stl`	Pocket core (shell, stays in)	1	50 x 92 x 91	
`core-outer-a-L.stl`	Pocket core (shell, stays in)	1	35 x 43 x 116	Shell core, stays in. Bolts to the bridge by its post.
`core-outer-b-L.stl`	Pocket core (shell, stays in)	1	35 x 60 x 116	Shell core, stays in. Bolts to the bridge by its post.
`core-rear-L.stl`	Pocket core (shell, stays in)	1	170 x 52 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-foot-b-R.stl`	Pocket core (shell, stays in)	1	68 x 100 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-foot-c-R.stl`	Pocket core (shell, stays in)	1	80 x 63 x 141	Shell core, stays in. Bolts to the bridge by its post.

File	What	Qty	Size (mm)	Notes
`core-knee-R-part1of2.stl`	Pocket core (shell, stays in)	1	50 x 92 x 40	Shell core, stays in. Bolts to the bridge by its post.
`core-knee-R-part2of2.stl`	Pocket core (shell, stays in)	1	50 x 92 x 91	
`core-outer-a-R.stl`	Pocket core (shell, stays in)	1	35 x 43 x 116	Shell core, stays in. Bolts to the bridge by its post.
`core-outer-b-R.stl`	Pocket core (shell, stays in)	1	35 x 60 x 116	Shell core, stays in. Bolts to the bridge by its post.
`core-rear-R.stl`	Pocket core (shell, stays in)	1	170 x 52 x 141	Shell core, stays in. Bolts to the bridge by its post.
`plug-socket-plug.stl`	Plug (removable)	4	41 x 41 x 15	Fills the open end of a socket liner; bolts to the floor.
`kit-spigot-bushing.stl`	Mould kit (reused)	2	79 x 79 x 22	Push fit in a 2-1/2 in hole-saw hole; flange on the far side.
`cast-in-base-rear-tube-plug-l.stl`	Cast-in part (stays in)	2	42 x 10 x 42	Cast in, stays.
`cast-in-base-rear-spigot-sleeve-l.stl`	Cast-in part (stays in)	2	44 x 69 x 44	Cast in, stays.
`cast-in-leg-rear-l-plug.stl`	Cast-in part (stays in)	2	30 x 30 x 10	Cast in, stays.
`cast-in-socket-liner-l1.stl`	Cast-in part (stays in)	4	45 x 45 x 80	Cast in, stays.

Base, front half

Folder `print/base-front/`, about 1061 g.

File	What	Qty	Size (mm)	Notes
`former-gutter-L-part1of2.stl`	Former (removable)	1	15 x 149 x 15	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-L-part2of2.stl`	Former (removable)	1	15 x 145 x 13	
`former-gutter-R-part1of2.stl`	Former (removable)	1	15 x 149 x 15	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-R-part2of2.stl`	Former (removable)	1	15 x 145 x 13	
`former-trough-part1of5.stl`	Former (removable)	1	116 x 35 x 19	Bolts to the floor from underneath (the base is cast top face down).
`former-trough-part2of5.stl`	Former (removable)	1	116 x 35 x 23	

File	What	Qty	Size (mm)	Notes
`former-trough-part3of5.stl`	Former (removable)	1	116 x 35 x 33	
`former-trough-part4of5.stl`	Former (removable)	1	116 x 35 x 24	
`former-trough-part5of5.stl`	Former (removable)	1	112 x 35 x 21	
`filler-corner-L.stl`	Filler (removable)	1	40 x 40 x 110	Rounded corner, and the plug that forms the draw-rod nut pocket; bolts to the floor and both walls.
`filler-corner-R.stl`	Filler (removable)	1	40 x 40 x 110	Rounded corner, and the plug that forms the draw-rod nut pocket; bolts to the floor and both walls.
`core-centre-part1of2.stl`	Pocket core (shell, stays in)	1	100 x 92 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-centre-part2of2.stl`	Pocket core (shell, stays in)	1	100 x 93 x 90	
`core-outboard-L.stl`	Pocket core (shell, stays in)	1	130 x 164 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-trough-band-L.stl`	Pocket core (shell, stays in)	1	101 x 55 x 91	Shell core, stays in. Bolts to the bridge by its post.
`core-draw-rod-L-part1of2.stl`	Pocket core (shell, stays in)	1	36 x 128 x 91	Shell core, stays in. Bolts to the bridge by its post.
`core-draw-rod-L-part2of2.stl`	Pocket core (shell, stays in)	1	37 x 128 x 40	
`core-foot-a-L.stl`	Pocket core (shell, stays in)	1	68 x 66 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-outboard-R.stl`	Pocket core (shell, stays in)	1	130 x 164 x 141	Shell core, stays in. Bolts to the bridge by its post.
`core-trough-band-R.stl`	Pocket core (shell, stays in)	1	101 x 55 x 91	Shell core, stays in. Bolts to the bridge by its post.
`core-draw-rod-R-part1of2.stl`	Pocket core (shell, stays in)	1	36 x 128 x 91	Shell core, stays in. Bolts to the bridge by its post.
`core-draw-rod-R-part2of2.stl`	Pocket core (shell, stays in)	1	37 x 128 x 40	
`core-foot-a-R.stl`	Pocket core (shell, stays in)	1	68 x 66 x 141	Shell core, stays in. Bolts to the bridge by its post.
`cast-in-base-front-tube-plug-l.stl`	Cast-in part (stays in)	2	42 x 10 x 42	Cast in, stays.
`cast-in-base-front-spigot-sleeve-l.stl`	Cast-in part (stays in)	2	44 x 69 x 44	Cast in, stays.

File	What	Qty	Size (mm)	Notes
`cast-in-leg-front-l-plug.stl`	Cast-in part (stays in)	2	30 x 30 x 10	Cast in, stays.

Uprights (two moulds: print quantities are for both)

Folder `print/upright/`, about 508 g.

File	What	Qty	Size (mm)	Notes
`kit-back-bracket.stl`	Mould kit (reused)	4	33 x 33 x 23	
`kit-front-bracket.stl`	Mould kit (reused)	8	50 x 50 x 22	Two at the top of the front board and two at its foot (only the side-plate bolt there).
`kit-floor-bracket.stl`	Mould kit (reused)	10	32 x 24 x 25	
`kit-saddle-flat.stl`	Mould kit (reused)	4	31 x 25 x 25	The bridge rests on them; its rods locate it.
`filler-toe-filler.stl`	Filler (removable)	2	80 x 60 x 48	Rounds the nose into the slope; the front board stands on it. Bolts to the floor.
`former-foot-gasket-groove-part1of4.stl`	Former (removable)	2	68 x 139 x 8	Pins into the floor.
`former-foot-gasket-groove-part2of4.stl`	Former (removable)	2	4 x 139 x 8	
`former-foot-gasket-groove-part3of4.stl`	Former (removable)	2	4 x 139 x 8	
`former-foot-gasket-groove-part4of4.stl`	Former (removable)	2	68 x 139 x 8	
`plug-slope-pocket--60.stl`	Plug (removable)	2	32 x 32 x 54	Forms a nut pocket in the slope; bolts to the front board; its rod stops inside it.
`plug-slope-pocket-40.stl`	Plug (removable)	2	32 x 32 x 54	Forms a nut pocket in the slope; bolts to the front board; its rod stops inside it.
`plug-tie-rod-pocket.stl`	Plug (removable)	4	32 x 32 x 49	Forms a top nut pocket: slides down its rod; the bridge holds it down.
`kit-plate-spacer.stl`	Mould kit (reused)	4	16 x 16 x 31	Between the bridge and the top plate, on its greased 3/8 bolt.
`kit-post-bushing.stl`	Mould kit (reused)	4	79 x 79 x 22	Push fit in a 2-1/2 in hole-saw hole in the floor, flange underneath.
`cast-in-post-l1-plug.stl`	Cast-in part (stays in)	4	30 x 30 x 10	Cast in, stays.

Beam

Folder `print/beam/`, about 255 g.

File	What	Qty	Size (mm)	Notes
`kit-corner-bracket.stl`	Mould kit (reused)	4	50 x 50 x 23	
`kit-floor-bracket.stl`	Mould kit (reused)	6	24 x 32 x 25	
`filler-back-edge-1-part1of4.stl`	Filler (removable)	1	136 x 40 x 40	Rounded back edge; bolts to the top wall.
`filler-back-edge-1-part2of4.stl`	Filler (removable)	1	136 x 40 x 40	
`filler-back-edge-1-part3of4.stl`	Filler (removable)	1	136 x 40 x 40	
`filler-back-edge-1-part4of4.stl`	Filler (removable)	1	132 x 40 x 40	
`kit-tube-locator.stl`	Mould kit (reused)	4	44 x 44 x 51	Square end into the tube, stem to the end wall, M4 x 70 through the wall into a nut slid in from inside the tube. Leaves a 16 mm hole in each beam end.
`kit-dowel-stop.stl`	Mould kit (reused)	2	18 x 18 x 11	Pushes into the dowel hole from outside so the dowel stands 10 mm proud.

Machine parts

Folder `print/machine/`, about 200 g.

File	What	Qty	Size (mm)	Notes
`part-pocket-cap-l3.stl`	Machine part	4	32 x 32 x 42	
`part-chain-mount-fixed.stl`	Machine part	1	25 x 38 x 21	
`part-chain-mount-moving.stl`	Machine part	1	37 x 123 x 45	
`part-sump-baffle-part1of2.stl`	Machine part	1	148 x 4 x 73	
`part-sump-baffle-part2of2.stl`	Machine part	1	148 x 4 x 73	
`part-sump-filter-frame-part1of2.stl`	Machine part	1	148 x 6 x 92	
`part-sump-filter-frame-part2of2.stl`	Machine part	1	148 x 6 x 92	